Section 6

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1 Final Comparison & Trade-off Discussion

1.1 Confusion Matrices

The confusion matrices provide a class-by-class view of the classification performance of the three models. The custom Model B shows substantial confusion between several visually similar pest classes, whereas the fine-tuned MobileNetV2 and EfficientNet-B0 models show predictions concentrated more strongly along the main diagonal. The matrices are presented in Figures 1a–1c.

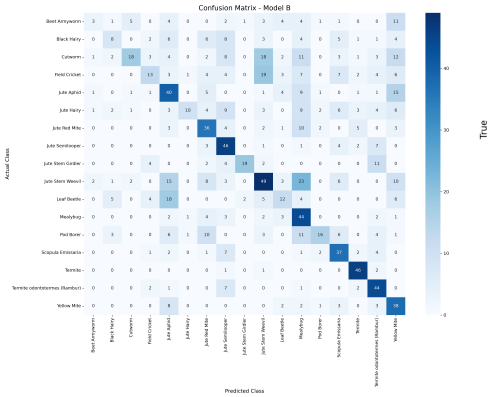
1.2 Quantitative Comparison of the Models

Table 1 summarises the reported classification metrics, memory footprint, computational cost, inference latency, and training duration of the three evaluated models.

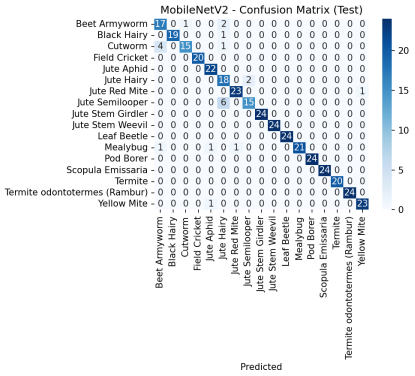
1.3 Trade-Off Analysis and Results Interpretation

The empirical results demonstrate a steep trade-off between representational capacity and physical edge deployability. Fine-tuning EfficientNet-B0 achieves a test accuracy of 97.63% and shows very little off-diagonal confusion across the 17 classes by adapting generic, pre-trained ImageNet representations. However, this accuracy advantage requires a substantially larger model and megabytes of runtime SRAM.

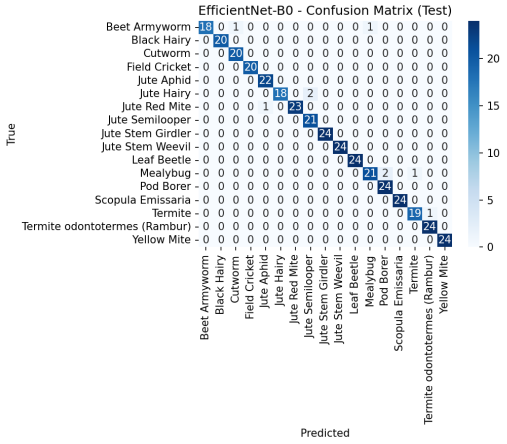
Conversely, Model B's custom sub-100k architecture uses depthwise separable convolutions to decouple spatial filtering from channel projection, reducing computational requirements. This design reduces storage to 16.30 KB in FP32 format, or 4.07 KB with INT8 quantization, and supports low inference latency. Nevertheless, training only 4,172 parameters from scratch imposes a severe capacity bottleneck. The Model B confusion matrix shows substantial confusion among morphologically similar pests, such as Beet Armyworm being misclassified as Cutworm or Leaf Beetle, and Jute Stem Weevil being confused with Mealybug. These results suggest that an ultra-compact model has difficulty learning deep hierarchical feature representations for fine-grained classification without large-scale pre-training.



(a) Confusion matrix for the custom Model B trained from scratch.



(b) Test confusion matrix for fine-tuned MobileNetV2.



(c) Test confusion matrix for fine-tuned EfficientNet-B0.

Figure 1: Test confusion matrices across the evaluated architectures.

Metric category	**Metric / dimension**	**Custom Model B (scratch)**	**Fine-tuned MobileNetV2**	**Fine-tuned EfficientNet-B0**
Accuracy	Test accuracy	44.11% (479/1,086)	94.20%	97.63%
Accuracy	Macro precision	0.4957	0.9452	0.9770
Accuracy	Macro recall	0.4468	0.9386	0.9755
Accuracy	Macro F1-score	0.4170	0.9390	0.9755
Memory footprint	Model size (disk/Flash)	16.30 KB (FP32); 4.07 KB (INT8)	8.80 MB	15.65 MB
Memory footprint	Peak working RAM (SRAM)	Minimal; below 100 KB activation buffers	Moderate/high; 2–6 MB tensor graph	High; 4–8 MB tensor graph
Memory footprint	MCU deployment feasibility	High; fits bare-metal Flash/SRAM	Infeasible without external Flash or an SBC	Infeasible without external Flash or an SBC
Computational cost	Total parameters	4,172 (below 100k)	2,245,649	4,029,325
Computational cost	Trainable parameters	4,172; all trained from scratch	21,777; classifier head only	21,777; classifier head only
Computational cost	CPU inference latency	Approx. 2.5–3.5 ms/image	33.28 ms/image	18.85 ms/image
Computational cost	Training duration	20 epochs (674.1 s)	5 epochs (621.3 s)	5 epochs (637.4 s)

Table 1: Quantitative comparison of Model B, MobileNetV2, and EfficientNet-B0 using the reported results.

1.4 Advantages

Table 2 compares the principal advantages of the custom Model B and the fine-tuned lightweight models.

1.5 Limitations of the Custom Sub-100k Model B Trained from Scratch

- **Severe representational capacity bottleneck:** With only 4,172 parameters and a maximum of 64 feature channels, the network has limited capacity to learn rich hierarchical features. Its reported test accuracy is 44.11% for the fine-grained 17-class classification task.
- **High data dependency and poor sample efficiency:** Random weight initialisation requires the model to learn basic spatial features, such as edges and textures, from scratch. With limited training data, this can lead to underfitting and confusion between visually similar pest classes.
- **Memory bandwidth bottlenecks on low-end hardware:** Although depthwise separable convolutions reduce theoretical multiply–accumulate operations, their low arithmetic intensity can cause memory-access stalls on microcontrollers without dedicated vector or SIMD acceleration.
- **High engineering and hyperparameter-tuning overhead:** Stable convergence may require repeated tuning of learning rates, channel dimensions, and regularisation settings, without guaranteeing production-level accuracy.

1.6 Limitations of Lightweight Pre-Trained Models

- **Large memory footprint for microcontrollers:** MobileNetV2 requires 8.80 MB of storage and EfficientNet-B0 requires 15.65 MB. These sizes generally exceed the internal Flash capacity of many bare-metal microcontrollers, requiring external memory or a single-board computer (SBC).
- **Higher peak working RAM demand:** Despite their parameter-efficient designs, the deep network topologies retain multi-channel intermediate activation tensors during inference, resulting in estimated peak working RAM requirements of 2–6 MB for MobileNetV2 and 4–8 MB for EfficientNet-B0.
- **Input-resolution mismatch:** These backbones are designed and pre-trained using standard image inputs. Heavily downscaled images can lose spatial information through early strided operations, potentially reducing the usefulness of learned features.
- **Higher computational latency on edge CPUs:** The reported CPU inference latencies are 33.28 ms/image for MobileNetV2 and 18.85 ms/image for EfficientNet-B0. Depending on the application and hardware, these

Dimension / category	**Custom sub-100k Model B**	**Fine-tuned lightweight models (MobileNetV2 / EfficientNet-B0)**
Predictive accuracy	Provides baseline competence on simpler features with very few parameters.	High accuracy and reliability; the reported test accuracy reaches 94.20% for MobileNetV2 and 97.63% for EfficientNet-B0, with few off-diagonal errors in their confusion matrices.
Storage footprint (Flash/ROM)	Ultra-compact: 16.30 KB in FP32, potentially fitting internal Flash on bare-metal microcontrollers such as STM32 and ESP32.	Compact compared with larger networks such as VGG or ResNet, but the reported model sizes are 8.80 MB and 15.65 MB.
Working memory (peak SRAM)	Shallow stages and small channel widths (16–64) keep intermediate activation buffers below 100 KB.	Inverted residual bottlenecks and depthwise separable blocks improve efficiency relative to conventional CNN designs, although runtime memory remains in the megabyte range.
Computational complexity	Depthwise separable convolutions separate spatial filtering and channel projection, helping minimise computation.	Compound scaling and architectural optimisations provide a balance between model capacity and computational cost.
Training efficiency and convergence	Provides full control over the network layers, loss components, and training process because the model is trained from scratch.	Rapid convergence and sample efficiency; high accuracy is reached after only five fine-tuning epochs by adapting pre-trained ImageNet representations.
Trainable parameter overhead	All 4,172 parameters can be directly modified, frozen, or pruned.	Fine-tuning updates only 21,777 parameters in the final classifier head, according to the reported configuration.
Hardware tailoring	Designed for edge hardware with a small architecture and simple ReLU activations.	Uses established architectural features, including inverted residuals and, in EfficientNet-B0, squeeze-and-excitation blocks.
Target deployment platform	Suitable for autonomous sensor nodes and low-power microcontrollers, subject to implementation and memory constraints.	Better suited to edge gateways and embedded Linux platforms such as Raspberry Pi or Jetson devices, or other systems with sufficient memory.

Table 2: Advantages of the custom model and the fine-tuned lightweight models.

delays may be unsuitable for strict real-time or battery-operated sensing requirements.

1.7 Practical Deployment Verdict

- **Bare-metal sensor nodes (e.g., ESP32 and STM32):** Model B is the most feasible of the three candidates under strict internal Flash and SRAM constraints because its model size is only 16.30 KB in FP32 format and 4.07 KB in INT8 format. However, its accuracy limitation must be considered. Possible improvements include grouping the 17 fine-grained pest classes into broader taxonomic families and applying INT8 quantisation, followed by evaluation to verify the resulting accuracy and memory use.
- **Edge gateways and intelligent vision platforms (e.g., Raspberry Pi and Jetson):** EfficientNet-B0 is the recommended option when sufficient memory and processing resources are available. It achieves the highest reported test accuracy (97.63%) and lower CPU inference latency (18.85 ms/image) than MobileNetV2 (33.28 ms/image), although its model storage requirement is higher at 15.65 MB compared with 8.80 MB for MobileNetV2.

1.8 Final Summary

Overall, the results show that no single model is optimal for every deployment scenario. Model B prioritises compactness, low memory use, and low inference latency, but its classification accuracy is insufficient for applications requiring reliable discrimination across all 17 pest classes. MobileNetV2 offers substantially better classification performance with a smaller storage footprint than EfficientNet-B0, while EfficientNet-B0 provides the highest accuracy and the lowest reported inference latency among the two fine-tuned models. Therefore, the final model should be selected by balancing classification requirements against the memory, processing, and deployment constraints of the target hardware.